EE4-57
EE9-CS2-2

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

MSc and EEE/EIE PART IV: MEng and ACGI

DISCRETE-EVENT SYSTEMS

Thursday, 6 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : D. Angeli
Second Marker(s) : E.C. Kerrigan

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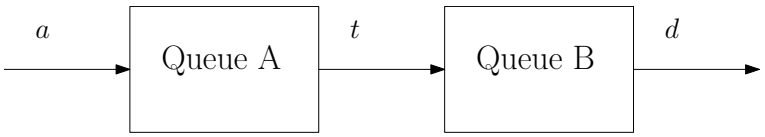


Figure 1.1 Queues operating in series

1. Consider 2 robots operating in series (see Fig. 1.1) and performing 2 different jobs, A and B respectively, on the pieces they process. Each time a piece arrives to the first robot, event a is recorded and a job of type A is scheduled and added to QUEUE A (of maximum length 2). Each time a job of type A is completed, a transfer event t is generated and the job leaves Queue A and is added to Queue B (of maximum length 2). Finally, each time a task of type B is completed, this is recorded as an event d.

   a) Define finite deterministic automata G_A and G_B to model the queues, arrivals, transfer and departures, which keep track of the number of tasks logged in each queue. [2]

   b) When a piece arrives at the first robot, but Queue A is already full while Queue B is empty, an exception event x is generated (instead of the corresponding a event). The event x does not modify the amount of jobs logged in Queue A, but adds 2 jobs to Queue B. This is meant to represent Robot B having to perform both types of operations on the piece. Modify automata G_A and G_B to allow for x events (and denote the new Automata as $\tilde{G}_A$, $\tilde{G}_B$ respectively). [2]

   c) Compute $\tilde{G}$, the parallel composition of $\tilde{G}_A$ and $\tilde{G}_B$. [4]

   d) Design a labelling automaton G_L, to discriminate between the situation in which event x has not occurred and the one in which x has occurred. [2]

   e) Compute the parallel composition $\tilde{G}||G_L$. [3]

   f) Assume that event x is unobservable, namely only the empty event ε is generated each time x occurs. Replace in $\tilde{G}||G_L$ event x with the empty event ε and denote by G_N the resulting non-deterministic automaton. [1]

   g) Build a diagnoser G_D that can determine if event x has occurred (YES / NO / MAYBE) by processing events of type a, t and d only. (For the sake of simplicity collapse all the diagnoser's states corresponding to a YES outcome into a single state). [6]

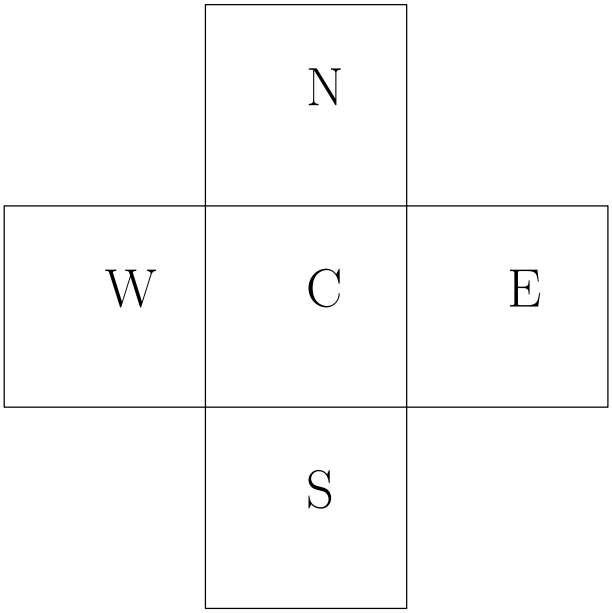


Figure 2.1 Arrangement of working cells

2. Two mobile robots move within three working cells. Robot 1 moves within the cells North (N), Center (C) and South (S) (see Fig.2.1), while Robot 2 moves within the cells West (W), Center (C) and East (E). Assume that initially Robot 1 is in cell N and Robot 2 is in cell W. Then, events of type s indicate that robot 1 is moving southwards by one cell, events of type n indicate movement northwards by 1, again. Similarly for Robot 2, events of type e indicate movement eastwards by 1 cell, and events of type w denote movement westwards.

   a) Build automata G_1 and G_2 modelling the movement of individual robots and their cell occupancy. [3]

   b) Compute the unsupervised behaviour by taking the parallel composition $G_1||G_2$. [3]

   c) Sketch the automaton G_S modelling the specification: "Robots should not be simultaneously in the same cell". [4]

   d) Assume that events n and e are uncontrollable (for instance due to high-priority). Is the specification G_S controllable? (justify your answer) [3]

   e) Design the supervisor and associated supervising automaton implementing the maximal controllable sub-language. [4]

   f) Assume next that the only terminal state is the initial state. Design a non-blocking supervising automaton implementing the maximal controllable sub-language. [3]

3. Consider the following scenario: a museum of finite capacity C_M, accepts visitors (arrival events, a) and lets them out once their visit is complete (departure events, d). While visitors are inside the museum they may decide to visit an exhibition (x event), but only up to capacity C_E (smaller than C_M). Once their visit of the exhibition is terminated they are let back in the museum (t events).

a) Model the above scenario by means of a Petri net (N_1), sketch its diagram and illustrate the physical meanining of its places, transitions and initial number of tokens associated to each place. [6]

b) Compute the incidence matrix C of the Petri Net. [2]

c) Compute, if any, the P-semiflows of the net and discuss their physical meaning. [3]

d) Add next a place to keep track of the total number of entrance tickets sold by the museum (one for each arrival). [2]

e) Add a transition to model that for every 1000 visitors the capacity of the museum might be increased by 1 (and the ticket counter decreased by 1000) without affecting the exhibition capacity. Denote this Petri Net by N_2. [2]

f) Is the modified Petri Net N_2 structurally bounded? Is the original Petri Net N_1 structurally bounded? (Justify your answer) [5]

4. An assembling machine is initially idle, waiting for 2 types of pieces (1 and 2) in order to operate. Arrival times of the 2 different pieces are exponentially distributed with rates λ_1 and λ_2, respectively. Assume that, once a piece of type 1 or 2 is available, its arrival process is interrupted (until assembling is completed) and the machine awaits for the complementary type of piece to arrive before completing the assembling operation. The latter is an event with exponentially distributed lifetime of rate λ_3. After completion of the assembling operation the machine is again in the idle state.

   a) Model the machine cycle with a Markov Chain, so as to keep track of the state of the machine, and the pieces available for the assembling operation. Sketch its transition diagram and write the corresponding Kolmogorov equations. [4]

   b) Is the Markov Chain ergodic? (Justify your answer) [2]

   c) Find the steady-state probability distribution and discuss how this can be used to compute the average fraction of time the arrival process of pieces of type 1 is interrupted. [4]

   d) Find the average duration of the machine operation cycle (Hint: model the cycle duration as the absorption time of a suitably modified Markov Chain). [4]

   e) Assume that a buffer of capacity 1 for pieces of type 1 is added at the machine, so that the arrival process of the corresponding type of pieces can continue during the assembling operation. Modify the Markov chain in order to take this extra degree of freedom into account. [3]

   f) For the modified Markov chain, compute the average fraction of time the arrival process of pieces of type 1 is interrupted. [3]